

AASHISH MALAKAR

ELECTRICAL ENGINEER - EV SYSTEMS & BATTERY SPECIALIST

PORTFOLIO LINK



- <https://drive.google.com/drive/folders/1fx93uuyA5d3PNDwBhJXviGaJwShRukol?usp=sharing>

CAREER OBJECTIVE

Electrical Engineer with 2+ years of experience in **lithium-ion battery systems**, **BMS integration**, and **DVP testing**. Proficient in **EV standards (ISO 26262, AIS-156)**, **SOC/SOH algorithms**, and **thermal management**. Demonstrated expertise in **designing, testing, and validating HEV/BEV systems**. Skilled in tools like **MATLAB**, **SolidWorks**, and **CANoe**. Passionate about delivering innovative solutions aligned with **sustainable mobility goals**.

KEY PROJECTS

Lithium-Ion Battery Pack Design (100 units of 72V, 60A)

- Successfully designed, developed, and tested 100 lithium-ion battery packs with specifications of 72V and 60A for EV applications. The project involved cell grading, BMS integration, thermal management, and ensuring compliance with safety standards.
- Set up lab vehicle for BEV prototype testing and supported EOL processes for quality assurance.

EV Powertrain Calibration

- Collaborated on the calibration of EV components, including inverters and on-board chargers, to enhance vehicle performance.

BMS Integration and Testing

- Designed and implemented BMS systems with **SOC/SOH estimation algorithms**.
- Validated battery packs for **DVP and homologation testing**, achieving successful compliance.

Arduino-Based Obstruction Sensors for Train

- Developed an Arduino-based system to enhance train safety by detecting obstacles on tracks and triggering preventive actions to avoid collisions or accidents.

PROFESSIONAL WORK EXPERIENCE

BMS Engineer

web inception | May 2023 - present

- Led the end-to-end design and production of lithium-ion battery packs, encompassing cell selection, grading, BMS integration, and capacity testing.
- Collaborated on **battery pack integration** projects, focusing on **thermal and electrical design feasibility**.
- Assisted in developing **wiring harness layouts** and system-level validation tests.
- Conducted **benchmarking and calibration** of EV subsystems, ensuring adherence to industry standards.

Intern

Swastik Poly_print pvt ltd | Jan 2023 - Apr 2023

- Conducted repair and testing procedures on **AC induction motors** and **Variable Frequency Drives (VFDs)**, acquiring **hands-on experience in troubleshooting** and **maintenance** of crucial components.

- Executed quality **testing for induction motors** and **variable frequency drives**.

KEY SKILLS

- **Battery Systems:** Lithium-ion battery pack design, BMS selection and integration, thermal management.
- **Control Algorithms:** SOC/SOH estimation, thermal control, safety optimization.
- **EV Standards:** ISO 26262 compliance, AIS-156, DVP preparation.
- **Tools & Software:** MATLAB, SolidWorks, ANSYS, CANoe, communication protocols (CAN, LIN).
- **Testing & Validation:** Battery pack testing, homologation testing, DVP and DFMEA.
- **System Design:** E-machine integration, wiring harness design, and calibration.

EDUCATION

- Bachelor of Engineering (B.E) - Electrical Engineering - 7.0 CGPA (July 2023)
Gujarat Technological University, GTU Palanpur, Gujarat
- HSC - Mathes - Guru Krupa Vidhya Sankul - (May 2019) - 71.78 %

CERTIFICATIONS

Electric Vehicle Technology Certification | Internshala (May 2023 - Jan 2024)

- **Trained in Electric Vehicle Technology, Engineering data analytics, Electrical & automotive engineering, Electronics systems and Embedded technology, Communication Protocols (CAN, LIN), Engineering business management, Battery & powertrain technology, Modeling & simulation, Dynamic analysis.**

SOFT SKILLS

- Problem-Solving & Analytical Thinking
- Team Collaboration & Leadership
- Effective Communication
- Time Management

PERSONAL INFORMATION

- **Date of Birth:** 24/04/2000
- **Languages Known:** Hindi, English, Gujarati
- **Marital Status:** Single
- Available for immediate joining.
- Willing to relocate for job opportunities

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